



BAHIR DAR INSTITUTE OF TECHNOLOGY
FACULTY OF MECHANICAL & INDUSTRIAL ENGINEERING
MANUFACTURING ENGINEERING CHAIR
ENGINEERING MATERIAL II COURSE GUIDE BOOK

ACADAMIC YEAR 2016

SEMESTER I

1. General Information

Course Title	Engineering Materials II					
Course Code	MEng3121				Instructor	
Pre requisite	Engineering Materials I				Name	Tewodros S.
Course type	Major/Compulsory/common				Office no	NDD
CP/ Cr.hr	<u>5</u>				Email	tedyo1312@gmail.com
Contact hours	Lecture	Tutorial	Lab	H. Study	Consultation hr	Friday 10:00 to 12:00 LT
	<u>2</u>	<u>2</u>	<u>1</u>	<u>5</u>	Lab Instructors /Assistants	
Target Group	Program/Department		Year	Section	Name	
	<u>Regular Mechanical engineering</u>		<u>3</u>	<u>A</u>	Office no	

2. Course Description

- Revision on phase diagrams,
- Concepts in phase transformations and Heat treatment,
- Introduction to ferrous metals, non-ferrous metals, organic and inorganic materials, corrosion and degradation of materials

3. Course objective/learning out come

At the end of the course, students will be able to:

- Understand the main concepts and kinetics of phase transformation, microstructural and property change in iron carbon alloy.
- Understand basic methods of iron and steel production, types of ferrous metals and effects of alloying elements.
- Explain Heat treatment, analyze applications of heat treatment
- Understand Non-ferrous metals, their engineering application, extraction and processing methods,
- Explain Ceramics materials, classification, properties and processing methods
- Explain polymer materials, classification, properties and processing methods
- Explain Composite materials, classification, properties and processing methods Explain corrosion and degradation in metals/materials

4. Course outline with teaching methodology, week and reference

Week	Course Contents	Teaching Methodology	Reference
Week 1	Chapter 1: Revision on Iron carbon Phase diagram	- Lecture - Group discussion	Text book and Reference book 1
Week 2	Chapter 1: Revision on Iron carbon Phase diagram	- Lecture - Group discussion - lab	Text book and Reference book 1,3
Week 3	Chapter 2: phase transformation and microstructural and property changes in iron–carbon alloys	- Lecture - Group discussion - lab	Text book and Reference book 2,3
Week 4	Chapter 2: phase transformation and microstructural and property changes in iron–carbon alloys	- Lecture - Group discussion - lab	Text book and Reference book 1,2
Week 5	Chapter 2: phase transformation and microstructural and property changes in iron–carbon alloys	- Lecture - Group discussion	Text book and Reference book 1,2
Week 6	Chapter 3: Iron and steel production, Applications and Processing of Metal Alloys; Heat treatments	- Lecture - Group discussion - lab	Text book and Reference book 1,3
Week 7	Chapter 3: Iron and steel production, Applications and Processing of Metal Alloys; Heat treatments	- Lecture - Group discussion - lab	Text book and Reference book 1,2,3
8	Mid Exam		
Week 9	Chapter 4: Structures , Properties , Applications and Processing of Ceramics	- Lecture - Group discussion	Text book and Reference book 1,2

Week 10	Chapter 4: Structures , Properties , Applications and Processing of Ceramics	- Lecture - Group discussion - lab	Text book and Reference book 1,3
Week 11	Chapter 5: Structures, Characteristics, Application Processing of Polymers	- Lecture - Group discussion	Text book and Reference book 1,2,3
Week 12	Chapter 5: Structures, Characteristics, Application Processing of Polymers	- Lecture - Group discussion	Text book and Reference book 1,2
Week 13	Chapter 6: Composites: Particle-Reinforced, Fiber-Reinforced, and Structural Composites	- Lecture - Group discussion - lab	Text book and Reference book 1,3
Week 14	Chapter 6: Composites: Particle-Reinforced, Fiber-Reinforced, and Structural Composites	- Lecture - Group discussion	Text book and Reference book 2,3
Week 15	Chapter 7: Corrosion and Degradation of Materials: Corrosion of Metals and Ceramics, Degradation of Polymers	- Lecture - Group discussion	Text book and Reference book 1,2
16	Final Exam		

5. Assessment type, weight and duration (*Minimum bincluding Mid-exam and Final-exam*)

Assessment type	Mark allotted	Duration
Type 1 quiz1	5%	Before Mid
Type 2 group ass.	10%	Before Mid
Mid exam	30%	
Type 1 quiz 2	5%	After Mid
Type 2 individual. Ass.	10%	After Mid
Final Exam	40%	
Total	100%	

6. Course Policy

All students are expected to abide by the code of conduct of students (article 166 and 166.1.1, of the Senate Legislation of Bahir Dar University May 20, 2005) throughout this course. Academic dishonesty, including cheating, fabrication, and plagiarism will not be tolerated and will be reported to concerned bodies for action.

Class attendance and participation: You are expected to **attend** class regularly. I will take attendance on regular days during the semester to ensure that students are coming to class, and if you miss class repeatedly, your grade will be affected as it has value. If you miss more than 85% lecture and tutorial and 100% for laboratory class attendance you will not sit for final exam.

7. Text and reference book

A. Text Book

- William D. Callister, “Material Science and Engineering” – An introduction, 7th edition, John Wiley & Sons, 2007

B. Reference Materials

1. Brain S. Mitchell, “An introduction to materials engineering and science for chemical and materials engineers”, 2004, Wiley Interscience.
2. Veron John, “Introduction to Engineering Materials,” Macmillan Education UK, May 9, 2003
3. Michael F. Ashby, Hugh Shercliff, David Cebon, Materials: Engineering, Science, Processing and Design, 3rd edition, Elsevier Ltd, 2014

8. Authorization

a. Prepared instructor's/s' Name: Tewodros Semeneh (MSc.)

Signature: _____ Date: 1/23/2024

b. Checked course chair's Name: _____

Signature: _____ Date: _____

c. Verified chair holder's Name: _____

Signature: _____ Date: _____

Faculty seal